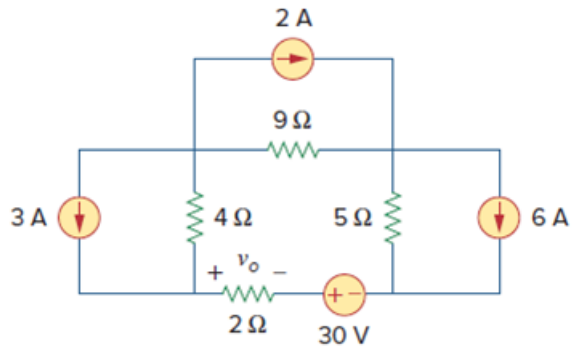


Problem

Obtain v_o in the circuit of Fig. 4.93 using source transformation. Check your result using *PSpice* or *MultiSim*.

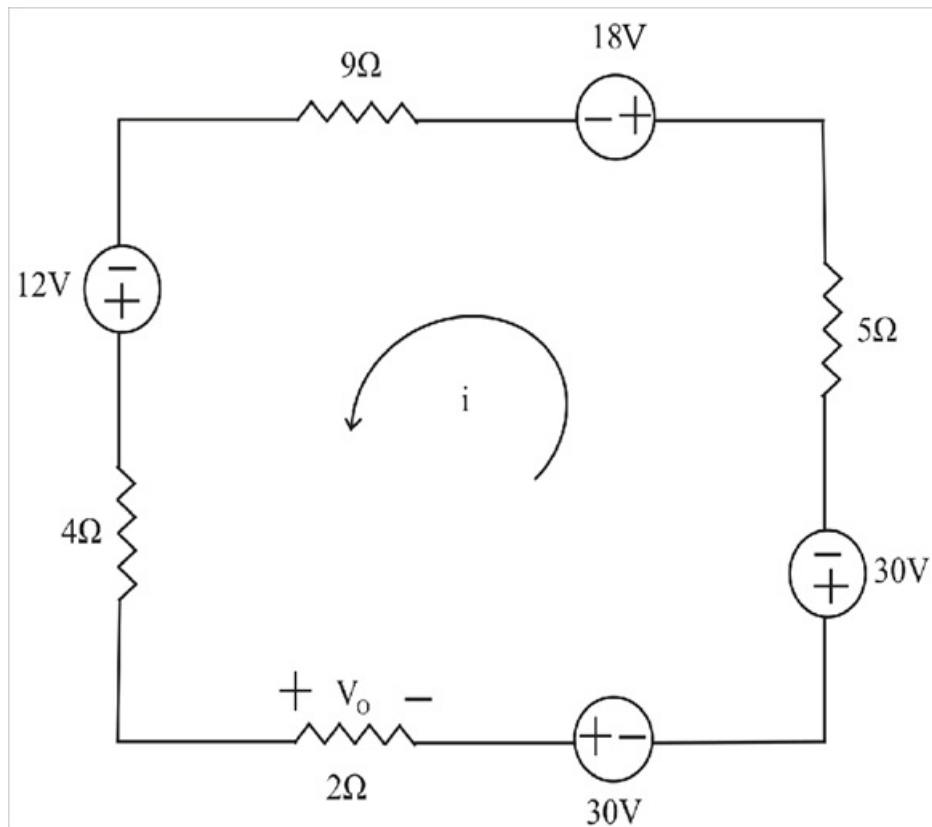
Figure. 4.93:



Step-by-step solution

Step 1 of 2

Transform only the current sources gives the circuit below.



Step 2 of 2

Applying KVL to the loop gives,

$$(4+9+5+2)i - 12 + 18 + 30 + 30 = 0 .$$

$$20i = -66 \quad \text{which leads to} \quad i = -3.3$$

$$V_0 = 2i = -6.6V .$$